t3-w05 — condition C0

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

> The curriculum objective alone. No lesson plan, no note.

INPUT LAYERS

Input layer	What it carries	Channel	In this condition
Scope	what is to be taught, and to whom	prompt	**present**
Prior knowledge	what the learner already knew coming in	prompt	absent
Materials	what the lesson was represented with	prompt	absent
Position	where this objective sat in the sequence	prompt	absent
Method	the shape of the teaching, as event labels	prompt	absent
Enacted content	what was actually explained and worked through	prompt	absent
Set work	what the learners were asked to do in class	prompt	absent
Assessment	the questions the lesson was checked with	prompt	absent
Figures	the drawings the lesson used	tool	absent

WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

```
Mathematics
```

`grade_level`

```
JSS1
```

`topic`

```
Three dimensional shapes II
```

`subtopic`

```
Three dimensional figures: properties of cylinders and spheres, and volume
```

`term`

```
Third Term
```

`week`

```
5
```

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

WHAT THE MODEL PRODUCED

Objective 1

Properties of Cylinders and Spheres

EXPLANATION

A **cylinder** has two identical circular faces (the bases) that are parallel to each other, and a curved surface that connects them. The distance between the bases is called the **height**. Key properties:
- Both bases have the same radius.
- The axis (a line through the centre of the bases) is perpendicular to the bases.
- The curved surface area is given by $2\pi r h$ where r is the radius of the base and h is the height.

A **sphere** is a perfectly round solid where every point on the surface is the same distance from the centre. Key properties:
- It has only one surface; there are no faces, edges or vertices.
- The distance from the centre to any point on the surface is the **radius** r.
- The surface area is $4\pi r^2$.

Both shapes are three-dimensional and their volume depends on the radius (and height for the cylinder).

WORKED_EXAMPLE

Worked Example: Identify the properties of a cylinder with radius 4 cm and height 10 cm.

- The bases are circles with radius 4 cm.
- There are two identical circular bases.
- The height (distance between the bases) is 10 cm.
- The axis is perpendicular to the bases.
- The curved surface area would be $2\pi \times 4 \times 10 = 80\pi$ cm².

Now, identify the properties of a sphere with radius 5 cm.

- It has a single continuous surface.
- Every point on the surface is 5 cm from the centre.
- No edges or vertices.
- The surface area would be $4\pi \times 5^2 = 100\pi$ cm².

1. A cylinder has a radius of 3 cm and a height of 7 cm. List its main properties.
answer: Two identical circular bases of radius 3 cm, height 7 cm, axis perpendicular to the bases, curved surface area $2\pi\times3\times7 = 42\pi$ cm².

2. A sphere has a radius of 6 cm. State its defining properties.
answer: A single continuous surface, every point 6 cm from the centre, no edges or vertices, surface area $4\pi\times6^2 = 144\pi$ cm².

3. Compare a cylinder and a sphere in terms of faces, edges, and vertices.
answer: A cylinder has 2 circular faces, 0 edges, 0 vertices (the curved surface is not a face). A sphere has 0 faces, 0 edges, 0 vertices.

Objective 2

Finding the Volume of Cubes and Cuboids

> EXPLANATION
> The volume of a solid tells us how much space it occupies. For rectangular solids the volume is found by multiplying the length, width and height.
>
> - **Cube**: all three edges are the same length (let's call it *a*). The volume is \(V = a \times a \times a = a^3\).
> - **Cuboid**: the edges can be different. If the length is *l*, the width is *w* and the height is *h*, then \(V = l \times w \times h\).
>
> Remember to keep the units consistent (e.g., all in centimetres) and write the final answer in cubic units (cm³, m³, etc.).

> WORKED_EXAMPLE
> **Example 1 – Cube**
>
> A cube has an edge length of 5 cm. What is its volume?
>
> Solution: \(V = a^3 = 5^3 = 5 \times 5 \times 5 = 125\) cm³.
>
> **Example 2 – Cuboid**
>
> A cuboid has length 8 cm, width 4 cm and height 3 cm. Find its volume.
>
> Solution: \(V = l \times w \times h = 8 \times 4 \times 3 = 96\) cm³.

1. A cube has an edge of 7 cm. Calculate its volume.
answer: 343 cm³

2. A cuboid measures 6 cm by 5 cm by 2 cm. What is its volume?
answer: 60 cm³

3. The length of a cuboid is twice its width, and its height is 3 cm. If the width is 4 cm, find the volume.
answer: 96 cm³

LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

Output layer	Made	Written from	Verdict
Title the reader-facing framing of the section	2	Scope	**grounded**
Explanation the principle, rule or relationship to review	2	nothing in the input	**unsourced**
Worked example a problem with its solution worked through	2	nothing in the input	**unsourced**
Figure a drawing the learner reads from	0	nothing in the input	not produced
Practice what the learner does	6	nothing in the input	**unsourced**
Answer what the learner checks against	6	nothing in the input	**unsourced**